

Mall Customers Data Analysis Report

1. Introduction

The objective of this project is to analyze customer data from a mall to understand spending behavior and identify patterns among customers. The dataset includes demographic and spending information.

2. Dataset Overview

The dataset contains 200 records with 5 features: CustomerID, Gender, Age, Annual Income, and Spending Score. There are no missing values.

3. Descriptive Analysis

Average Age: 38.85, Average Income: 60.56k, Average Spending Score: 50.2. Income shows highest variability among features.

4. Data Visualization Insights

Age is slightly right skewed. Income is positively skewed. Spending score is fairly balanced. Few outliers exist in income.

5. Gender Analysis

Females spend slightly more than males (51.53 vs 48.51). Males have slightly higher income (62.23k vs 59.25k).

6. Correlation Analysis

No strong relationship between income and spending. Age shows a negative relationship with spending score.

7. Inferential Statistics

T-test shows p-value = 0.4137, so there is no statistically significant difference in spending between genders.

8. Confidence Interval

Mean spending score = 50.2. 95% CI = (46.62, 53.78). We are 95% confident true mean lies in this range.

9. Business Insights

Focus on customer segmentation based on spending behavior. Gender is not a strong factor. Younger customers tend to spend more.

10. Conclusion

Spending behavior is driven by patterns rather than gender or income. Segmentation-based marketing is recommended.